Tools and APIs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Recall and choose your starting point

> Record your starting explanation for Tools and APIs before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Conversation State and Memory

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Retrieval Augmented Generation

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Source Quality

Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Tools and APIs trace and record the first step you can explain without help.

Use W2: Trace the worked case for Tools and **API contract**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Tools and APIs explanation against the supplied facts before discussing it.

Use the worked case for Tools and **API contract**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Tool accepts room_id string. Calls are {room_id:7}, {room_id:K} and {room_id:K, delete:true}. Validate; K is absent from the registry.

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Tools and APIs target to a trace and a justified conclusion.

A tool can calculate or search supplied notes. The request is 'What is 17×6, and where do roots take up water?' Assign each subtask and describe **output** checks.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Tools and APIs case and show the deciding evidence.

An assistant has calculator and source-search tools. Request: 12 times 9 and habitat definition from a class pack.

Choose and verify tools.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A tool can look up book availability and also place an order. The user asks only whether a book is available. Which action is authorized?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 21 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Tools and APIs adds to Verified Study Buddy and keep the evidence.

Allow only retrieval and calculator tools with validated **argument**s and visible results.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A tool can calculate or search supplied notes. The request is 'What is 17×6, and where do roots take up water?' Assign each subtask and describe **output** checks.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Tools and APIs facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied Tools and APIs facts and keep your reasoning.

Proposed call lookup_room has room_id=B and extra **argument** book=true. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Audit the limit

> Audit the limit. Apply the supplied Tools and APIs facts and keep your reasoning.

A report omits this qualification: A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Tools and APIs facts and keep your reasoning.

Tool accepts room_id string. Calls are {room_id:7}, {room_id:K} and {room_id:K, delete:true}. Validate; K is absent from the registry.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Tools and APIs facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Tools and APIs

> Check a library lookup call before it reaches a tool. Explain the labelled step before checking the changed condition.

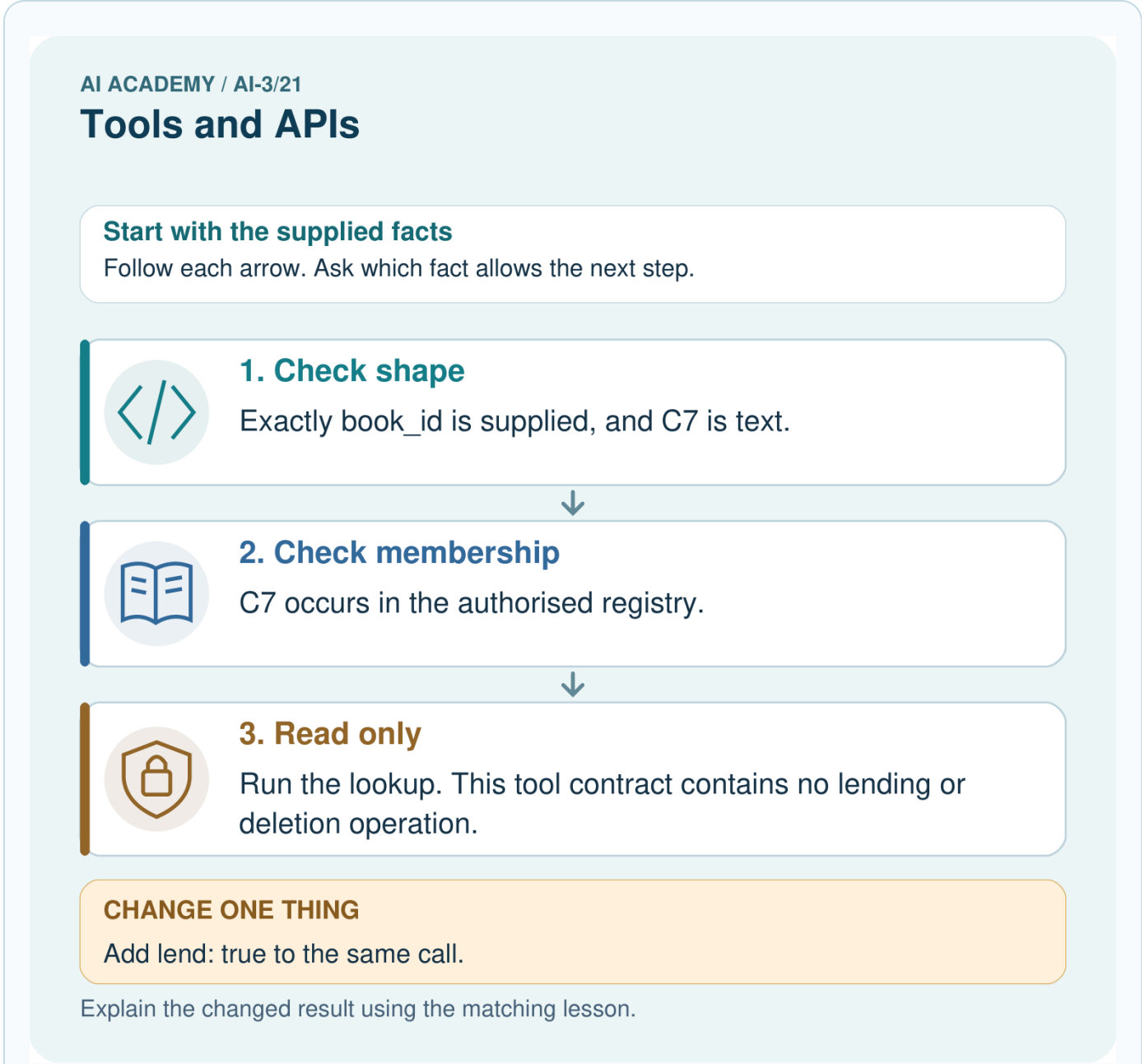
Check a library lookup call before it reaches a tool.

lookup_book accepts exactly one field: book_id, a string. Registry IDs are C7 and M2. The proposed call is {book_id: C7}.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/21

Tools and APIs



Follow the labelled picture with the matching explanation. Open the vector diagram

Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Check a library lookup call before it reaches a tool.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Run the lookup. This tool contract contains no lending or deletion operation.

1. Check shape

2. Read only

3. Check membership

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Read only Run the lookup. This tool contract contains no lending or deletion operation. Read the picture in order. Run the lookup.

This tool contract contains no lending or deletion operation. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add lend: true to the same call. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Reject the call before invoking the tool.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reject the call before invoking the tool. An extra operation is outside the exact allowed contract, even when book_id is valid.

Repair a missing building block

Review Conversation State and Memory (ai-3/20). Goal: Explain the **mechanism** and terminology of conversation state and retention. Short practice: What is selected_book after the correction and after exit?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the **mechanism** and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Source Quality (ai-3/15). Goal: Explain the **mechanism** and terminology of conflicting and time-sensitive sources. Short practice: Which source fields must be checked before answering day 9?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

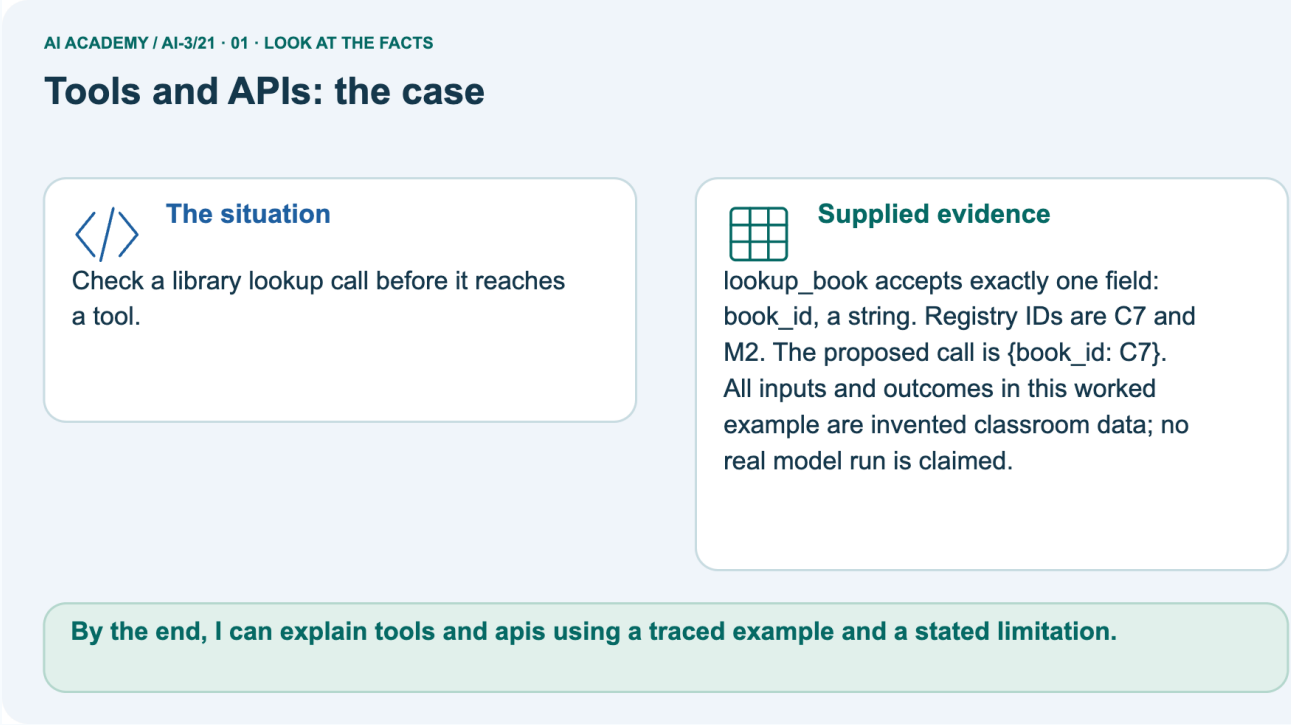
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Tools and APIs picture from its labels, then record a relevant limit.

01. Tools and APIs: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

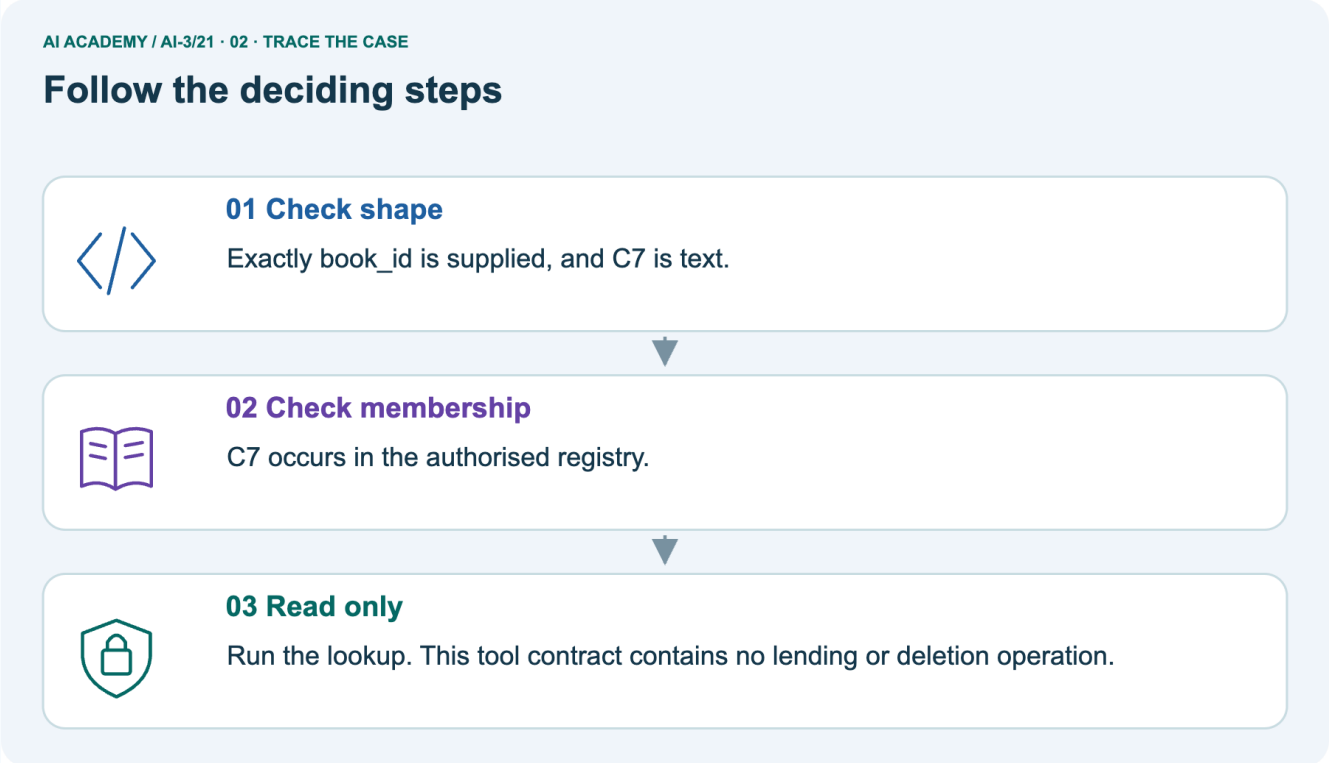
Check a library lookup call before it reaches a tool. Supplied facts: lookup_book accepts exactly one field: book_id, a string. Registry IDs are C7 and M2. The proposed call is {book_id: C7}.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



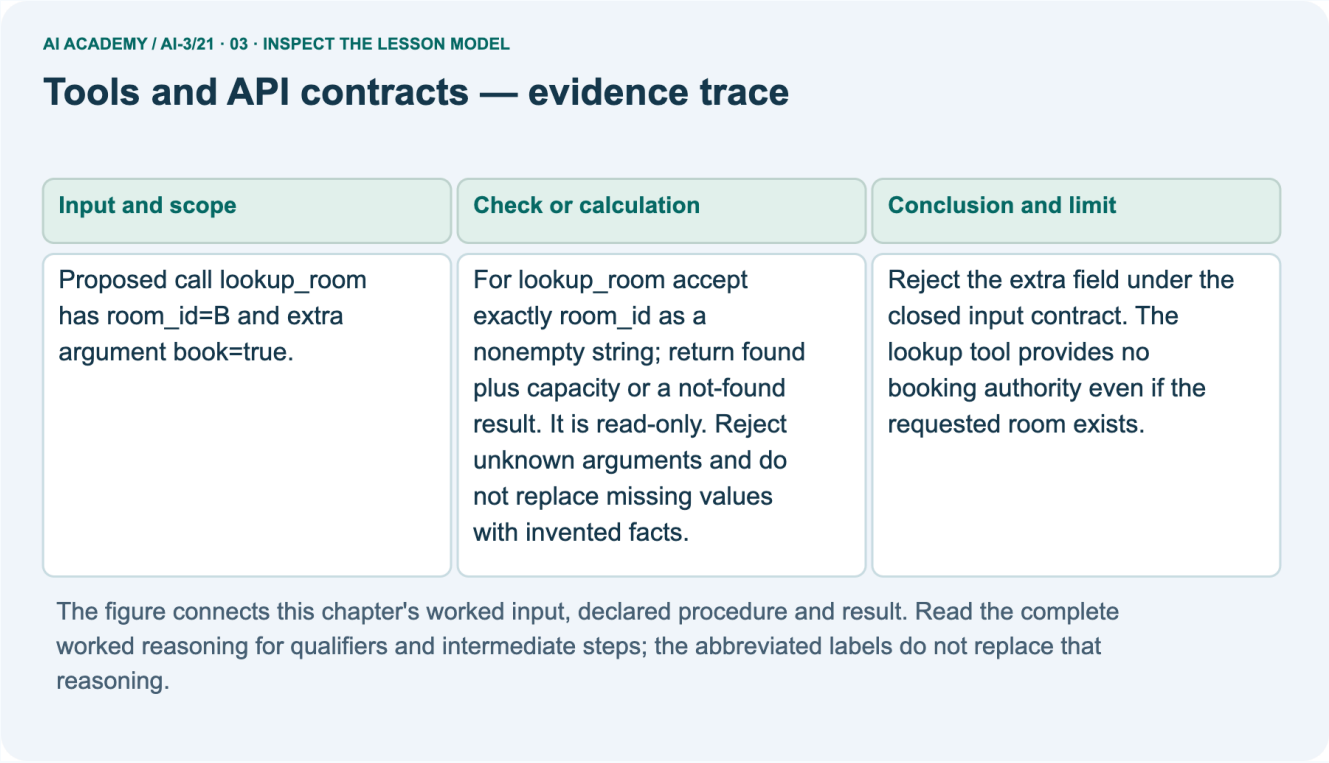
Follow the labelled picture with the matching explanation. Open the vector diagram

Check shape: Exactly book_id is supplied, and C7 is text. → Check membership: C7 occurs in the authorised registry. → Read only: Run the lookup. This tool contract contains no lending or deletion operation.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Tools and API contracts — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Tools and **API contract**s — **evidence** trace. Proposed call lookup_room has room_id=B and extra **argument** book=true. | For lookup_room accept exactly room_id as a nonempty string; return found plus capacity or a not-found result.

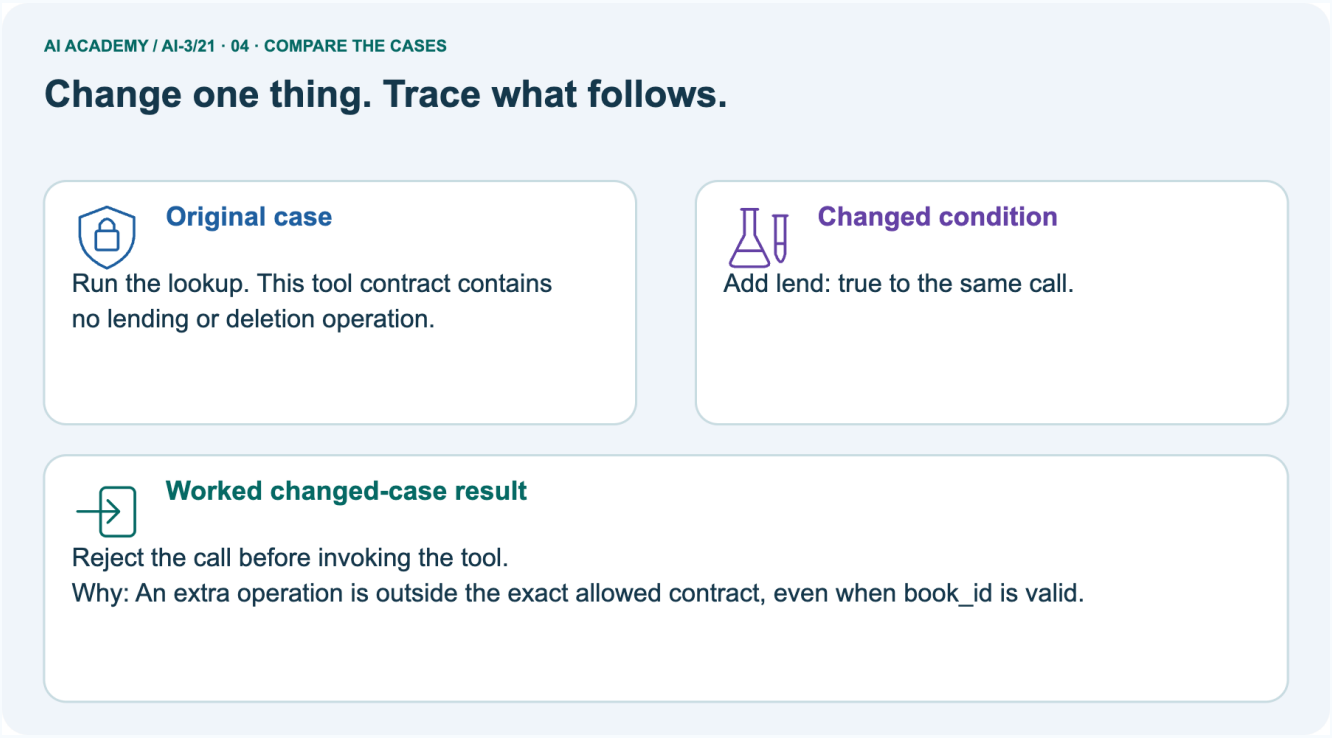
It is read-only. Reject unknown **argument**s and do not replace missing values with invented facts. | Reject the extra field under the closed **input** contract.

The lookup tool provides no booking authority even if the requested room exists.

> Read the labels and values. Explain the relationship the model shows.

> **My picture explanation, deciding evidence and one limit**

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

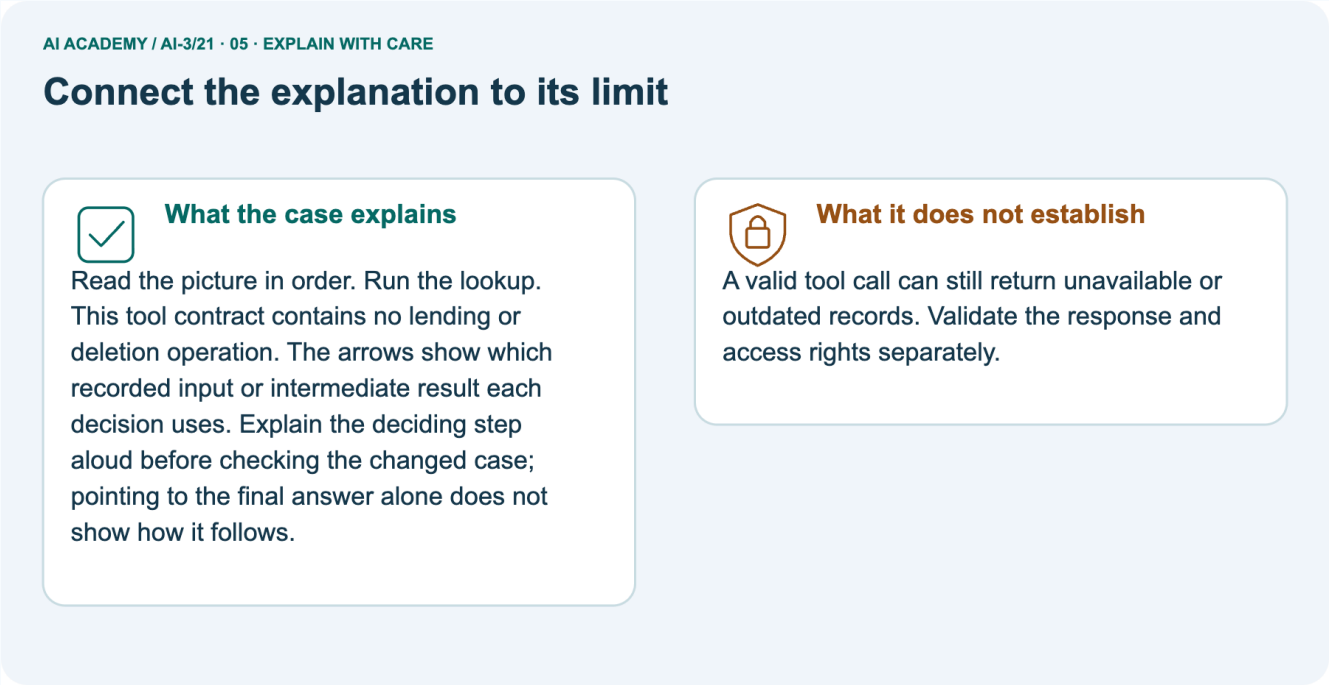
Original case: Run the lookup. This tool contract contains no lending or deletion operation. Changed condition: Add lend: true to the same call. Expected result: Reject the call before invoking the tool.

Reason: An extra operation is outside the exact allowed contract, even when book_id is valid.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Run the lookup. This tool contract contains no lending or deletion operation. The arrows show which recorded **input** or intermediate result each decision uses.

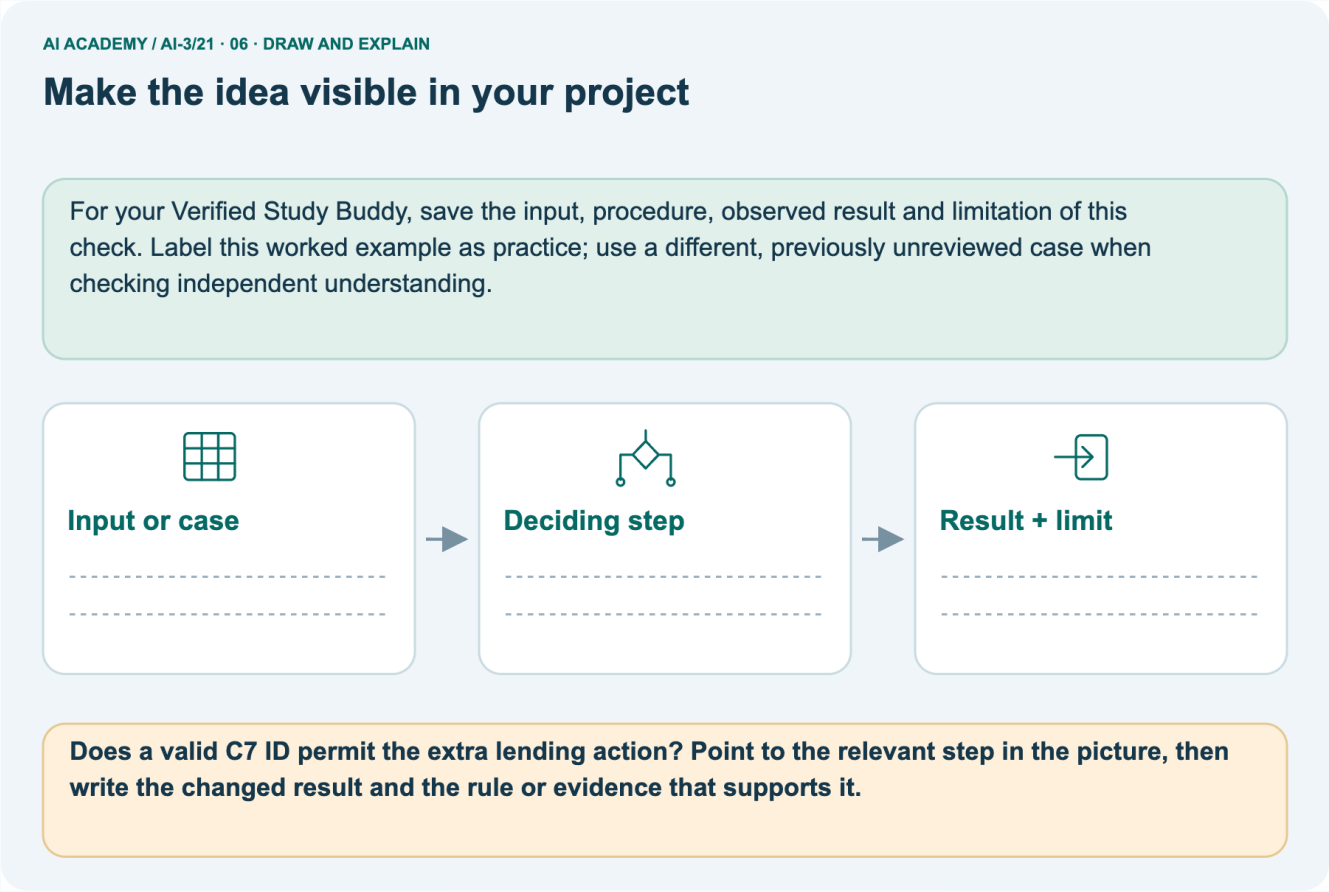
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: A valid **tool call** can still return unavailable or outdated records.

Validate the response and access rights separately.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.